

Exercise Sheet 10 – Solutions

Linear Algebra

What do the terms mean?

- **Coordinate vector** $[v]_B$: “How much of each basis vector do I need to build v ?” If $v = \alpha b_1 + \beta b_2 + \dots$, then $[v]_B = (\alpha, \beta, \dots)$.
- **Representation matrix** $[T]_B$: The columns are the images of the basis vectors, expressed *in the basis* B . Column $j = [T(b_j)]_B$.
- **Standard basis** E : $e_1 = (1, 0, \dots)$, $e_2 = (0, 1, \dots)$, ... In this basis $[v]_E = v$ (you read off the coordinates directly).
- **Change-of-basis matrix** P : columns = the basis vectors of B in standard coordinates. Then $[v]_E = P[v]_B$ and conversely $[v]_B = P^{-1}[v]_E$.
- **Change of basis for transformations**: $[T]_E = P[T]_B P^{-1}$ resp. $[T]_B = P^{-1}[T]_E P$.

Exercise 39

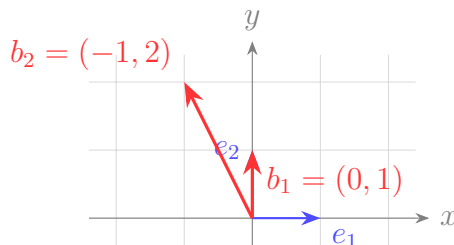
Problem. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (x + 2y, -3x)$. Basis $B = \{(0, 1), (-1, 2)\}$, vector $v = (1, 2)$.

We name the basis vectors: $b_1 = (0, 1)$, $b_2 = (-1, 2)$.

What needs to be done?

- **(a)** $[T]_B$ in two ways: (1) directly – images $T(b_1), T(b_2)$ in B -coordinates as columns; (2) first $[T]_E$, then change of basis with P .
- **(b)** $[T(v)]_B$ in two ways: (1) first $[v]_B$, then via $[T]_B[v]_B$; (2) first $T(v)$ in standard, then change of basis.

[Graphical explanation – not part of the solution]



The basis B (red) is “rotated” against the standard basis (blue). b_1 lies on the y -axis, b_2 points diagonally up-left. $[v]_B$ asks: how much b_1 and how much b_2 do I need to reach v ?

(a) Compute $[T]_B$

Method 1: Directly from the definition.

Step 1: Compute the images of the basis vectors.

$$\begin{aligned}T(b_1) &= T(0, 1) = (0 + 2 \cdot 1, -3 \cdot 0) = (2, 0) \\T(b_2) &= T(-1, 2) = (-1 + 2 \cdot 2, -3 \cdot (-1)) = (3, 3)\end{aligned}$$

Step 2: Convert each image into B -coordinates.

Why? The columns of $[T]_B$ are the images – but *expressed in the basis B* . So for each image we have to find out: “How much b_1 and how much b_2 do I need to build it?”

The general setup (set it up once, then reuse it again and again): a vector (p, q) is to be written as $\alpha b_1 + \beta b_2$. Substitute and multiply out:

$$(p, q) = \alpha(0, 1) + \beta(-1, 2) = \underbrace{(0 \cdot \alpha - 1 \cdot \beta)}_{\text{1st component}}, \underbrace{(1 \cdot \alpha + 2 \cdot \beta)}_{\text{2nd component}} = (-\beta, \alpha + 2\beta).$$

Comparing components gives two equations:

$$p = -\beta \quad \text{and} \quad q = \alpha + 2\beta.$$

We now solve these for each image separately.

(a) $T(b_1) = (2, 0)$, so $p = 2, q = 0$:

• From $p = -\beta$: $2 = -\beta \Rightarrow \beta = -2$.

• From $q = \alpha + 2\beta$: $0 = \alpha + 2(-2) = \alpha - 4 \Rightarrow \alpha = 4$.

Check: $4 \cdot (0, 1) + (-2) \cdot (-1, 2) = (0, 4) + (2, -4) = (2, 0) \checkmark$. So $[T(b_1)]_B = (4, -2)$.

(b) $T(b_2) = (3, 3)$, so $p = 3, q = 3$:

• From $p = -\beta$: $3 = -\beta \Rightarrow \beta = -3$.

• From $q = \alpha + 2\beta$: $3 = \alpha + 2(-3) = \alpha - 6 \Rightarrow \alpha = 9$.

Check: $9 \cdot (0, 1) + (-3) \cdot (-1, 2) = (0 + 3, 9 - 6) = (3, 3) \checkmark$. So $[T(b_2)]_B = (9, -3)$.

Step 3: Assemble the columns.

$$[T]_B = \begin{pmatrix} 4 & 9 \\ -2 & -3 \end{pmatrix}.$$

Method 2: First $[T]_E$, then change of basis.

Step 1: Set up $[T]_E$ (columns = $T(e_1), T(e_2)$ in standard):

$$T(e_1) = T(1, 0) = (1, -3), \quad T(e_2) = T(0, 1) = (2, 0) \Rightarrow [T]_E = \begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}.$$

Step 2: Change-of-basis matrix P and its inverse. Columns of P = the B -vectors:

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}, \quad \det P = 0 \cdot 2 - (-1) \cdot 1 = 1, \quad P^{-1} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}.$$

[Explanation added] The inverse of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$: swap the main diagonal, negate the off-diagonal, divide by the determinant. Here $\det P = 1$, so the division drops out.

Step 3: Compute $[T]_B = P^{-1}[T]_E P$. We compute from the right: first $[T]_E P$, then P^{-1} in front.

(a) $[T]_E P$ (row times column, entry by entry):

$$\begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot (-1) + 2 \cdot 2 \\ -3 \cdot 0 + 0 \cdot 1 & -3 \cdot (-1) + 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}.$$

(b) P^{-1} in front:

$$\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2 + 1 \cdot 0 & 2 \cdot 3 + 1 \cdot 3 \\ -1 \cdot 2 + 0 \cdot 0 & -1 \cdot 3 + 0 \cdot 3 \end{pmatrix} = \begin{pmatrix} 4 & 9 \\ -2 & -3 \end{pmatrix}.$$

Both methods: $[T]_B = \begin{pmatrix} 4 & 9 \\ -2 & -3 \end{pmatrix}$.

(b) **Compute** $[T(v)]_B$, $v = (1, 2)$

Method 1: First $[v]_B$, **then** $[T]_B[v]_B$.

Step 1: Convert $v = (1, 2)$ **into** B -**coordinates**. Same setup as in (a): $(p, q) = \alpha b_1 + \beta b_2$ leads to $p = -\beta$ and $q = \alpha + 2\beta$. Here $p = 1$, $q = 2$:

- From $p = -\beta$: $1 = -\beta \Rightarrow \beta = -1$.
- From $q = \alpha + 2\beta$: $2 = \alpha + 2(-1) = \alpha - 2 \Rightarrow \alpha = 4$.

Check: $4(0, 1) + (-1)(-1, 2) = (0 + 1, 4 - 2) = (1, 2) \checkmark$. So $[v]_B = (4, -1)$.

Step 2: Using the theorem $[T(v)]_B = [T]_B[v]_B$.

[Explanation added] Instead of recomputing $T(v)$, we use: in B -coordinates, applying T is simply “matrix $[T]_B$ times coordinate vector $[v]_B$ ”. That is the whole point of the representation matrix.

$$[T(v)]_B = \begin{pmatrix} 4 & 9 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \cdot 4 + 9 \cdot (-1) \\ -2 \cdot 4 + (-3) \cdot (-1) \end{pmatrix} = \begin{pmatrix} 16 - 9 \\ -8 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -5 \end{pmatrix}.$$

Method 2: First $T(v)$ **in standard**, **then change of basis**.

Step 1: $T(v)$ directly. $T(1, 2) = (1 + 2 \cdot 2, -3 \cdot 1) = (5, -3)$, so $[T(v)]_E = (5, -3)$.

Step 2: Into B -**coordinates with** P^{-1} .

$$[T(v)]_B = P^{-1}[T(v)]_E = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ -3 \end{pmatrix} = \begin{pmatrix} 10 - 3 \\ -5 + 0 \end{pmatrix} = \begin{pmatrix} 7 \\ -5 \end{pmatrix}.$$

Both methods: $[T(v)]_B = (7, -5)$.

[Graphical explanation – not part of the solution]

$$\begin{array}{ccc} [v]_B = (4, -1) & \xrightarrow{\cdot [T]_B \text{ (Method 1)}} & [T(v)]_B = (7, -5) \\ \uparrow [\cdot]_B & & \uparrow P^{-1} \\ v = (1, 2) & \xrightarrow{T \text{ direct (Method 2)}} & T(v) = (5, -3) \end{array}$$

Both methods lead to the same corner at the top right – the diagram “commutes”. Method 1 goes around the top (first into B -coordinates, then $[T]_B$), Method 2 goes around the bottom (first T in standard, then convert with P^{-1}).

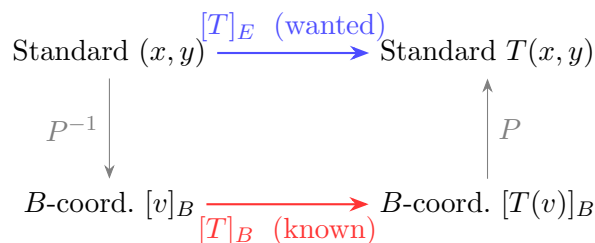
Exercise 40

Problem. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $[T]_B = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, basis $B = \{(1, 2), (1, 3)\}$. Wanted: explicit formula for $T(x, y)$.

What needs to be done?

- We know $[T]_B$, but want T in normal coordinates. Plan: compute $[T]_E = P [T]_B P^{-1}$ ($P = B$ -vectors as columns). From $[T]_E$ we read off the formula directly.

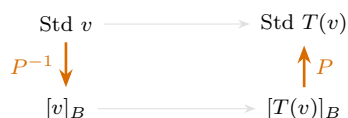
[Graphical explanation – not part of the solution]



$[T]_E$ (top, wanted) is obtained as a detour: first go down into the B -world with P^{-1} , apply the known matrix $[T]_B$ there, then go back up with P . That is exactly $[T]_E = P [T]_B P^{-1}$ – read right to left like the arrows.

Step 1: Change-of-basis matrix P and inverse.

[Orientation in the square – not part of the solution]

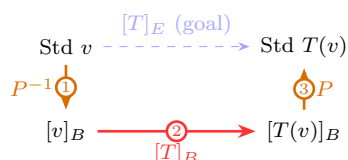


We are here: we only build the two vertical arrows P and P^{-1} – the “translators” between the standard and B worlds. We don’t touch the horizontal arrows yet.

$$P = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}, \quad \det P = 3 - 2 = 1, \quad P^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}.$$

Step 2: Compute $[T]_E = P [T]_B P^{-1}$.

[Orientation in the square – not part of the solution]



We are here: now we walk the detour around the square – basically in a circle from top left to top right. (1) go down into the B -world with P^{-1} , (2) apply the known matrix $[T]_B$ at the bottom, (3) go back up with P . This whole path is the wanted top arrow. We compute in exactly this order: first $[T]_B P^{-1}$ (arrows 1+2), then P in front (arrow 3).

First $[T]_B P^{-1}$:

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 3-4 & -1+2 \\ 0+2 & 0-1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}.$$

Then P in front:

$$[T]_E = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -1+2 & 1-1 \\ -2+6 & 2-3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 4 & -1 \end{pmatrix}.$$

Step 3: Read off the formula.

[Orientation in the square – not part of the solution]

$$\begin{array}{ccc} \text{Std } v & \xrightarrow{[T]_E \text{ (now known)}} & \text{Std } T(v) \\ \downarrow & & \uparrow \\ [v]_B & \xrightarrow{\quad\quad\quad} & [T(v)]_B \end{array}$$

We have arrived: the top arrow is done, $[T]_E = \begin{pmatrix} 1 & 0 \\ 4 & -1 \end{pmatrix}$. Because in the standard basis $[v]_E = v$, you read the formula directly off this matrix.

The matrix $[T]_E$ applied to (x, y) :

$$\begin{pmatrix} 1 & 0 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 4x - y \end{pmatrix}.$$

[Explanation added] In the standard basis $[v]_E = v$ (you read off the coordinates directly). Therefore $T(x, y) = [T]_E \cdot (x, y)^\top$ – you may simply apply the matrix to (x, y) and get the formula directly, without any further conversion.

$$T(x, y) = (x, 4x - y).$$

Check: $T(1, 2) = (1, 4 - 2) = (1, 2) = b_1$ and $T(1, 3) = (1, 4 - 3) = (1, 1)$. In B -coordinates: $b_1 = (1, 0)_B$ (1st column of $[T]_B$ ✓) and $(1, 1) = 2b_1 - b_2$, i.e. $(2, -1)_B$ (2nd column ✓).

Exercise 41

Problem. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x-2z, y-3z)$. Bases $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ (in the domain) and $C = \{(1, 0), (1, 1)\}$ (in the codomain).

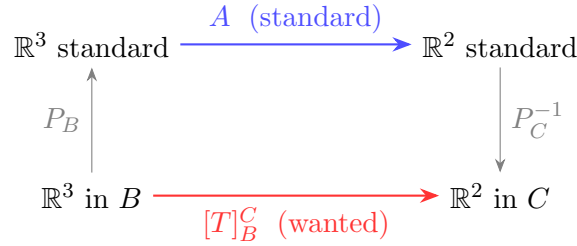
We name: $b_1 = (1, 0, 0)$, $b_2 = (1, 1, 0)$, $b_3 = (1, 1, 1)$, $c_1 = (1, 0)$, $c_2 = (1, 1)$.

What do the terms mean?

- $[T]_B^C$: transformation from basis B (source) to basis C (target). Column $j = [T(b_j)]_C$.

- Here **two** different bases are involved – one for the source, one for the target.

[Graphical explanation – not part of the solution]



Two spaces, two “languages” each. $[T]_B^C$ (bottom) leads from B directly to C . The detour around the top: with P_B to standard ($\mathbb{R}^3$), map with A , with P_C^{-1} into the C -language ($\mathbb{R}^2$). Hence $[T]_B^C = P_C^{-1} A P_B$.

(a) Compute $[T]_B^C$

Method 1: Directly from the definition.

Step 1: Images of the B -vectors.

$$T(b_1) = T(1, 0, 0) = (1 - 0, 0 - 0) = (1, 0)$$

$$T(b_2) = T(1, 1, 0) = (1 - 0, 1 - 0) = (1, 1)$$

$$T(b_3) = T(1, 1, 1) = (1 - 2, 1 - 3) = (-1, -2)$$

Step 2: Each image in C -coordinates. Write $(p, q) = \alpha c_1 + \beta c_2 = \alpha(1, 0) + \beta(1, 1) = (\alpha + \beta, \beta)$. From this: $\beta = q, \alpha = p - q$.

- $T(b_1) = (1, 0)$: $\beta = 0, \alpha = 1 \Rightarrow [T(b_1)]_C = (1, 0)$.
- $T(b_2) = (1, 1)$: $\beta = 1, \alpha = 0 \Rightarrow [T(b_2)]_C = (0, 1)$.
- $T(b_3) = (-1, -2)$: $\beta = -2, \alpha = -1 - (-2) = 1 \Rightarrow [T(b_3)]_C = (1, -2)$.

Step 3: Assemble the columns.

$$[T]_B^C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

Method 2: First standard bases, then change of basis.

Step 1: Matrix A of T in standard bases (columns = $T(e_1), T(e_2), T(e_3)$):

$$T(e_1) = (1, 0), T(e_2) = (0, 1), T(e_3) = (-2, -3) \Rightarrow A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \end{pmatrix}.$$

Step 2: Change-of-basis matrices.

$$P_B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \text{ (B-vectors as columns)}, \quad P_C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad P_C^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Step 3: Formula $[T]_B^C = P_C^{-1} A P_B$. First AP_B :

$$AP_B = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \end{pmatrix}.$$

Then P_C^{-1} in front:

$$[T]_B^C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

Both methods: $[T]_B^C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$

(b) $[v]_B$ and $[T(v)]_C$ for $v = (x, y, z)$ and verification

Step 1: $[v]_B$. $(x, y, z) = \alpha b_1 + \beta b_2 + \gamma b_3 = (\alpha + \beta + \gamma, \beta + \gamma, \gamma)$:

$$\gamma = z; \quad \beta + \gamma = y \Rightarrow \beta = y - z; \quad \alpha + \beta + \gamma = x \Rightarrow \alpha = x - y.$$

$$[v]_B = (x - y, y - z, z).$$

Step 2: $[T(v)]_C$. First $T(v) = (x - 2z, y - 3z)$. In C -coordinates ($\beta = q, \alpha = p - q$):

$$\beta = y - 3z; \quad \alpha = (x - 2z) - (y - 3z) = x - y + z.$$

$$[T(v)]_C = (x - y + z, y - 3z).$$

Step 3: Verify $[T]_B^C[v]_B = [T(v)]_C$.

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} x - y \\ y - z \\ z \end{pmatrix} = \begin{pmatrix} (x - y) + z \\ (y - z) - 2z \end{pmatrix} = \begin{pmatrix} x - y + z \\ y - 3z \end{pmatrix}. \checkmark$$

$[v]_B = (x - y, y - z, z), \quad [T(v)]_C = (x - y + z, y - 3z), \quad \text{theorem confirmed. } \checkmark$

Exercise 42

Problem. $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ with $T(X) = AX - XA$, where $A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$. Standard basis $E = \{E_{11}, E_{12}, E_{21}, E_{22}\}$.

- (a) Compute $[T]_E$.
- (b) Using $[T]_E$: basis and dimension of $\text{Ker}(T)$ and $\text{Im}(T)$.

What do the terms mean?

- E_{ij} is the matrix with a 1 at position (i, j) and zeros elsewhere. $\dim \mathbb{R}^{2 \times 2} = 4$.
- **Coordinates of a matrix** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in E : simply (a, b, c, d) – the four entries in order.
- Column j of $[T]_E$ = coordinates of $T(j\text{-th basis matrix})$.

[Graphical explanation – not part of the solution]

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \xrightarrow{\text{coordinates}} (a, b, c, d)$$

2×2 matrix

vector in $\mathbb{R}^4$

The trick: a 2×2 matrix is “in truth” just a vector with 4 entries. Through these glasses ($\mathbb{R}^{2 \times 2} \cong \mathbb{R}^4$) T becomes a perfectly ordinary 4×4 matrix, and you compute kernel/image as always with Gauss.

(a) Compute $[T]_E$

Step 1: Apply T to each basis matrix. We need $T(E_{ij}) = AE_{ij} - E_{ij}A$.

(a) $T(E_{11})$:

$$AE_{11} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}, \quad E_{11}A = \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} \Rightarrow T(E_{11}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \hat{=} (0, 1, 1, 0).$$

(b) $T(E_{12})$:

$$AE_{12} = \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix}, \quad E_{12}A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow T(E_{12}) = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \hat{=} (-1, 2, 0, 1).$$

(c) $T(E_{21})$:

$$AE_{21} = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{21}A = \begin{pmatrix} 0 & 0 \\ 2 & -1 \end{pmatrix} \Rightarrow T(E_{21}) = \begin{pmatrix} -1 & 0 \\ -2 & 1 \end{pmatrix} \hat{=} (-1, 0, -2, 1).$$

(d) $T(E_{22})$:

$$AE_{22} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, \quad E_{22}A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \Rightarrow T(E_{22}) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \hat{=} (0, -1, -1, 0).$$

Step 2: Coordinate vectors as columns.

$$[T]_E = \begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & 2 & 0 & -1 \\ 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

(b) Kernel and image from $[T]_E$

Step 1: Bring $[T]_E$ to row echelon form.

$$\begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & 2 & 0 & -1 \\ 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -1 & -1 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow[R_4 \rightarrow R_4 + R_2]{R_3 \rightarrow R_3 - 2R_2} \begin{pmatrix} \boxed{1} & 2 & 0 & -1 \\ 0 & \boxed{-1} & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivots in columns 1 and 2 $\Rightarrow$ rank = 2.

Step 2: Dimensions.

$$\dim \text{Im}(T) = \text{rank} = 2, \quad \dim \text{Ker}(T) = 4 - 2 = 2.$$

Step 3: Determine the kernel. Variables (w_1, w_2, w_3, w_4) stand for $X = w_1 E_{11} + w_2 E_{12} + w_3 E_{21} + w_4 E_{22}$. The row echelon form gives:

$$\text{Row 2: } -w_2 - w_3 = 0 \Rightarrow w_2 = -w_3$$

$$\text{Row 1: } w_1 + 2w_2 - w_4 = 0 \Rightarrow w_1 = -2w_2 + w_4 = 2w_3 + w_4$$

Free variables: w_3, w_4 .

$$(w_1, w_2, w_3, w_4) = w_3(2, -1, 1, 0) + w_4(1, 0, 0, 1).$$

Translated back into matrices:

$$w_3\text{-part} \hat{=} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = A, \quad w_4\text{-part} \hat{=} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

This makes intuitive sense: $T(A) = A^2 - A^2 = 0$ and $T(I) = AI - IA = A - A = 0$. Both really lie in the kernel.

$$\text{Basis of Ker}(T): \left\{ \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}. \quad \dim \text{Ker}(T) = 2.$$

Step 4: Determine the image. Pivots are in columns 1 and 2 $\Rightarrow$ the corresponding original images $T(E_{11})$ and $T(E_{12})$ form a basis of the image.

[Explanation added] Why the *original* columns and not those of the echelon form? During Gaussian elimination the columns change – but the pivot *positions* reveal which of the original columns are linearly independent. These (here columns 1 and 2, i.e. $T(E_{11})$ and $T(E_{12})$) form the basis of the image. Their coordinates $(0, 1, 1, 0)$ and $(-1, 2, 0, 1)$ are translated back into matrices.

$$\text{Basis of Im}(T): \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \right\}. \quad \dim \text{Im}(T) = 2.$$

Check: $\dim \text{Ker} + \dim \text{Im} = 2 + 2 = 4 = \dim \mathbb{R}^{2 \times 2}$. ✓